

Ligjerata 11d

Ekuacionet e Puasonit

Të proceset adiabatike nuk kemi shprehim të matësirë me rrethime:

$$dQ = 0$$

Nga P.I.T

$$dQ = du + dA$$

$$dA = -du / \int$$

$$\boxed{A = -\int du}$$

Ekuacionet i parë Puasonit

Udhërrimi adiabatik i gjendjes së gazit ideal.

$$dQ = du + dA, \text{ pra } dA = p dV$$

$$dQ = du + p dV$$

$$dQ = 0$$

$$du = -p dV, du = C_v dT$$

$$pV = nRT, n = 1 \text{ mol}$$

$$pV = RT$$

$$p = \frac{RT}{V}$$

$$C_p = C_v + nR$$

$$n = 1 \text{ mol}$$

$$du + p dV = 0$$

$$R = C_p - C_v$$

$$C_v dT + \frac{RT}{V} dV = 0$$

$$C_v dT + \frac{C_p - C_v}{V} T dV = 0 \quad / : C_v \cdot T$$

$$\frac{dT}{T} + \frac{C_p - C_v}{C_v} \cdot \frac{dV}{V} = 0$$

$$\frac{dT}{T} + \left(\frac{C_p}{C_v} - 1 \right) \cdot \frac{dV}{V} = 0$$

$$\frac{C_p}{C_v} = \kappa$$

$$\frac{dT}{T} + (\kappa - 1) \frac{dV}{V} = 0 \quad / \int$$

$$\int_{T_1}^{T_2} \frac{dT}{T} + (\kappa - 1) \int_{V_1}^{V_2} \frac{dV}{V} = 0$$

ose

$$(\kappa - 1) \int_{V_1}^{V_2} \frac{dV}{V} = - \int_{T_1}^{T_2} \frac{dT}{T}$$

$$(\kappa - 1) \left(\ln V \Big|_{V_1}^{V_2} \right) = - \left(\ln T \Big|_{T_1}^{T_2} \right)$$

$$(\kappa - 1) (\ln V_2 - \ln V_1) = (\ln T_2 - \ln T_1)$$

$$(\kappa - 1) \ln \frac{V_2}{V_1} = - \ln \frac{T_2}{T_1} = - \left(- \ln \frac{T_1}{T_2} \right)$$

$$(\kappa - 1) \ln \frac{V_2}{V_1} = \ln \frac{T_1}{T_2}$$

$$\left(\frac{V_2}{V_1} \right)^{\kappa - 1} = \frac{T_1}{T_2}$$

$$\boxed{T_1 V_1^{\kappa - 1} = T_2 V_2^{\kappa - 1}} \quad *$$

$$p_1 \cdot V_1 = RT_1 \Rightarrow T_1 = \frac{p_1 V_1}{R}$$

$$p_2 V_2 = RT_2 \Rightarrow T_2 = \frac{p_2 V_2}{R}$$

$$\frac{p_1 V_1}{R} V_1^{\kappa - 1} = \frac{p_2 V_2}{R} V_2^{\kappa - 1}$$

$$p_1 V_1^{\kappa} = p_2 V_2^{\kappa}$$

$$p_1 V_1^{\kappa - 1 + 1} = p_2 V_2^{\kappa - 1 + 1}$$

$$p_1 V_1^{\kappa} = p_2 V_2^{\kappa} = \text{konst}$$

$$p V^{\kappa} = \text{konst} \quad \text{Eku I i Puasonit}$$

